

Inception Time Mamba Network with Distribution Balanced Loss for Multi-Label ECG Classification

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ARTICLE INFO

Keywords:

ECG Classification

Multi-Label

Inception Time

Mamba

Distribution-Balanced Loss

Imbalance.

ABSTRACT

The 12-lead electrocardiogram (ECG) is a standard and effective tool for diagnosing cardiovascular diseases. Despite its widespread use, existing ECG classification models often encounter challenges such as high computational demands and imbalanced datasets. To address these issues, we propose the Inception Time Mamba Network (ITMN), a novel deep neural architecture that integrates Inception and Mamba Blocks. This hybrid design leverages the Inception Block's flexible kernel sizes for local ECG feature extraction, while the Mamba Block captures global dependencies—significantly improving noise suppression and feature refinement, both essential for accurate multi-label ECG classification. Additionally, we incorporate a Distribution-Balanced loss function to mitigate the commonly overlooked issue of data imbalance in ECG analysis. Our lightweight model, comprising only 0.43 million parameters, achieves state-of-the-art performance on the PTB-XL and CPSC 2018 datasets, attaining an AUC of 97.63% and a TPR of 92.37%. These results underscore the model's strong potential for deployment in real-world, resource-constrained IoT environments.

1. Introduction

As reported by the World Health Organization, ischemic heart disease is the leading cause of mortality worldwide, accounting for over 17.8 million deaths in 2017 [1]. Remarkably, 80% of heart attacks are preventable with early pre-diagnosis using electrocardiogram signals (ECGs) [2]. Electrocardiograms, collected using noninvasive tools, are widely used in clinics for the early diagnosis of heart diseases. Cardiac abnormalities can be detected by evaluating the heart's electrical activity, either from a standard 12-lead system as used in hospitals or from a 3-lead system in some wearable devices. Traditional hospital ECG devices have several drawbacks: they require multiple sensors, which can cause discomfort; ECG annotation is time-consuming and labor-intensive; and they cannot provide continuous monitoring as they are only used during hospital visits. In contrast, wearable ECG devices offer a practical solution for continuous recording and early diagnosis. Advances in wearable technology and the Internet of Things (IoT) have spurred the development of these portable devices [3, 4]. Moreover, recent breakthroughs in smart medicine have facilitated the automation of ECG-based arrhythmia identification, aiding clinicians. However, enhancing the performance of automatic arrhythmia classification systems remains a significant challenge.

The standard 12-lead ECG, which includes leads I, II, III, aVR, aVL, aVF, and V1 to V6, captures the heart's electrical activity from various spatial angles and is a widely used non-invasive diagnostic tool for cardiac disorders. Clinicians detect these disorders by examining changes in the QRS complex, P-wave, T-wave, and ST-segment across the 12 leads [5]. However, accurately classifying a 12-lead ECG, especially in patients with multiple cardiac disorders, poses significant challenges even for experienced clinicians. This underscores the urgent need for automated ECG classification methods that can detect co-occurring cardiac conditions. Beyond the inherent complexity of interpreting 12-lead data, several specific challenges complicate automated analysis. 12-Lead Data Complexity arises from the sheer volume and intricate interplay of information across multiple channels, where patterns at various temporal scales (e.g., distinct waveforms like QRS, P, T) must be simultaneously considered. Furthermore, accurately modeling Temporal Dependencies is critical, as cardiac events unfold sequentially, requiring algorithms to capture

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both short-term dynamics and long-range relationships within the signal. ECG signals are also highly susceptible to various interferences, making Noise Robustness a significant concern; algorithms must effectively distinguish true physiological signals from artifacts. Lastly, real-world clinical datasets often suffer from severe Data Imbalance, where rare but critical cardiac conditions are significantly underrepresented, posing a major hurdle for models to learn effectively across all classes. While most current research focuses on single-label ECG classification, which associates one ECG record with one disorder [6, 7], the importance of multi-label ECG classification is increasing. This approach, where one record corresponds to multiple concurrent disorders [8, 9], addresses the high prevalence of multi-morbidity among cardiovascular disease patients and its correlation with increased mortality risk.

Over the past decade, machine learning algorithms have been widely applied to classify ECG signals. Techniques such as Support Vector Machines (SVM), Principal Component Analysis (PCA), K-Nearest Neighbors (KNN), and Hidden Markov Models (HMM) have played significant roles in the automatic detection of arrhythmias, thereby improving diagnostic accuracy [5, 7, 10, 11, 12, 13, 14]. Traditional machine learning approaches typically follow a three-stage pipeline: preprocessing, feature extraction, and classification. Among these stages, feature extraction is particularly crucial. However, manually crafted features often limit overall performance and accuracy. In recent years, deep learning—particularly artificial neural networks—has achieved impressive results in domains such as image processing, natural language processing, signal processing, and computer vision. This success has led to growing interest in applying deep learning techniques to ECG signal analysis. Deep neural networks are especially effective at automatic feature extraction, making them more accurate than traditional methods for arrhythmia detection. Various deep learning architectures have been proposed for ECG classification, including Convolutional Neural Networks (CNNs) [15, 16, 17, 18, 19, 20, 21, 22, 23], Long Short-Term Memory (LSTM) networks [24, 25, 26], hybrid CNN-LSTM models [27, 9, 28, 29, 30, 31, 32], Binary Neural Networks [33], and CNN-Transformer architectures [34, 35, 36, 37]. These models have significantly improved the accuracy of automatic arrhythmia classification and serve as valuable tools for assisting cardiologists. Despite their success, most existing models utilize all 12 ECG leads directly, often neglecting lead-specific information and the inherent complexity of inter-lead relationships. This poses challenges for practical deployment, especially in resource-constrained environments such as IoT-based healthcare devices. Moreover, training deep learning models for multi-label ECG classification is inherently difficult due to the exponential growth in the number of possible label combinations and the severe class imbalance commonly found in real-world datasets. These factors complicate the process of determining optimal classification thresholds and often result in suboptimal model performance.

To address the existing limitations in effectively utilizing the comprehensive information and diverse characteristics of 12-lead multi-label ECG signals for improved classification accuracy, we propose a lightweight ITM Block architecture. This architecture integrates the complementary strengths of the Inception Block and the Mamba Block. Specifically, the Inception Block is designed to extract detailed local ECG features, such as the precise positions of the P, T, and U waves, as well as the PR, QT, and ST intervals. In contrast, the Mamba Block focuses on capturing global ECG characteristics, including the morphology of the QRS complex, RR intervals, and wave amplitudes. By combining these two blocks, our network effectively leverages both local and global features, resulting in higher classification accuracy compared to existing approaches. The main contributions of this work are as follows:

1. The Inception Time Mamba (ITM) Block: We propose the ITM Block, a hybrid architecture that combines the Inception Time [38] and Mamba [39] blocks. This integration exploits the multi-scale capabilities inherent in the Inception Time block for capturing localized ECG features while harnessing the global feature extraction proficiency of the Mamba block. The ITM block is designed to enhance noise suppression and feature refinement, which is critical for achieving accurate multi-label ECG classification.
2. Distribution Balanced Loss (DB Loss): We suggest the utilization of DB Loss to address the data imbalance challenges often overlooked by previous methods. Our findings reveal the effectiveness of DB Loss in handling ECG signals, where a single data type can encompass multiple diseases with varied morphologies.
3. Inception Time Mamba Network (ITMN): We propose the ITMN, constructed using the ITM Block and integrating the Distributed Balanced Loss. Despite its compact architecture, it effectively addresses the multi-class ECG classification problem.
4. Experiments on Multi-label ECG Datasets: Our tests conducted on two multi-label ECG datasets demonstrate that our network significantly outperforms contemporary state-of-the-art models in various classification tasks.

The remainder of this paper is organized as follows. Section 2 presents related works in the field of ECG classification. Section 3 details our ITM model incorporated with DB Loss. Section 4 covers the datasets, model evaluation methods

and experimental settings. Section 5 is about experimental results and discussion. Finally, Section 6 summarizes the contributions and suggests future research directions.

2. Related Works

Classifying ECG signals with deep learning is challenging yet meaningful, with many studies focusing on single-label ECG classification. Few studies address multi-label ECG classification. This section explores traditional and deep learning algorithms for ECG classification.

2.1. Machine Learning-based methods

In traditional ECG classification, machine learning begins with the extraction of morphological and statistical features, such as the width and amplitude of QRS waves [5], along with those derived from wavelet transforms or hidden Markov models [11]. The subsequent step involves classification using algorithms like multilayer perceptrons, K-nearest neighbors, principal component analysis, and support vector machines [12]. For example, discrete wavelet transformer coefficients and a fully connected layer have been employed for feature extraction and classification in [13]. Similarly, Pham et al. [7] utilized higher-order spectral analysis and a random forest for these tasks, while Sun et al. [14] developed an ensemble classifier utilizing 169 diverse features. These traditional methods, however, depend heavily on the accurate detection of ECG fiducial points and skilled feature extraction from complex cardiac signals, which are both challenging and limit performance. Consequently, there is a growing shift towards applying deep learning networks to ECG classification, which addresses the limitations of conventional machine learning approaches.

2.2. Deep Learning-based methods

Recent advancements in deep learning have led to the development of specialized Convolutional Neural Network (CNN) architectures for ECG classification. Wang et al. [18] introduced a 33-layer CNN that incorporates non-local convolutional blocks and attention mechanisms, enabling the model to capture spatial and channel-wise features as well as long-range dependencies in ECG signals. Several studies have evaluated state-of-the-art architectures, including Fully Convolutional Networks [19, 40] achieving an AUC of approximately 85.43%, 1D Residual Networks [40, 41] reaching an AUC of 86.04%, and InceptionTime [38] attaining an AUC of 86.66%. Similarly, Xiong et al. [20] proposed a densely connected CNN designed to enhance the modeling of inter-lead correlations in single-lead ECGs for myocardial infarction detection. Cheng et al. [21] developed a ResCNN architecture tailored for multi-label ECG arrhythmia classification. Jia et al. [22] introduced an ensemble method that combines sequence generation and multitask learning modules to enhance the classification of rhythm abnormalities. Tran et al. [23] proposed an enhanced GhostNet architecture that integrates advanced components such as the Channel Shuffle Module, Ghost Module, and Efficient Channel Attention to improve classification accuracy. While these models demonstrate strong performance on ECG classification tasks, they often fail to address the problem of class imbalance. The study on the MVMSNet architecture [17] utilizes multi-view learning and multi-scale CNNs with coordinate attention mechanisms to boost feature extraction. This model achieved an AUC of 89.1%, demonstrating its effectiveness in capturing diverse ECG patterns. Although Yang et al.'s approach [17] incorporated multi-scale convolutional blocks, coordinate attention, and knowledge distillation for improved efficiency, it still did not address the persistent issue of class imbalance in ECG datasets. In summary, these methods employ a wide range of techniques for multi-label ECG classification, often leveraging the multi-view characteristics of multi-lead ECG data to enhance predictive performance. However, many of them tend to overlook critical challenges such as network topology complexity and data imbalance, and their performance still falls short of the accuracy levels required for real-world clinical applications.

To leverage the complementary strengths of spatial and temporal modeling, numerous studies have explored hybrid CNN–LSTM architectures for ECG classification. ATI-CNN [9] integrates VGG-inspired convolutional layers, LSTM units, and attention mechanisms to enhance spatial fusion across ECG leads. The model achieves an AUC of 83.54% and an F1-score of 23.22, indicating moderate performance with improved spatial representation. ECGNet [27] adopts a CNN–LSTM framework in which the LSTM module is applied first to model temporal dependencies. It reports an AUC of 83.30 and an F1-score of 27.24, demonstrating its ability to learn sequential features effectively. ACNet [31] consists of five convolutional blocks, followed by a bidirectional recurrent neural network (RNN), an attention mechanism, and a fully connected output layer. Although specific evaluation metrics are not provided, the architecture is designed to learn both spatial and temporal characteristics of ECG signals. Further developments include dual-attention deep neural networks that combine CNNs, BiLSTMs, and attention modules to enhance interpretability [30, 31]. In addition,

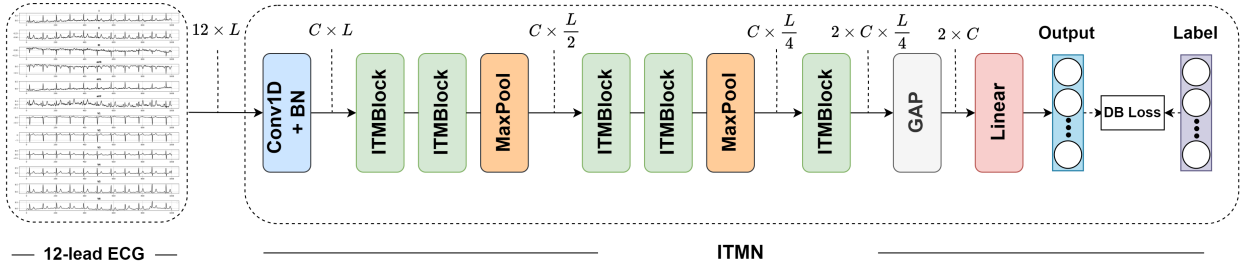


Figure 1: Overview of the Proposed Method for ECG Multi-Label Classification. The input 12-lead ECG signals are processed directly using the Inception Time Mamba Network (ITMN). Initially, feature extraction involves passing the input data through one 1D convolutional layer, followed by five successive ITMBlocks and two max-pooling layers. Subsequently, a linear layer is employed to perform multi-label classification, generating predictions corresponding to the input signals. These predicted outputs are then compared against the ground truth labels using the Distribution-Balanced Loss (DB Loss) function. This loss computation facilitates updating the network's weights throughout the training process, effectively addressing the issue of data imbalance.

Ganeshkumar et al. [32] proposed an explainable model that integrates CNNs with Grad-CAM to visualize and interpret ECG abnormalities. More recently, Alamatsaz et al. [28] introduced a lightweight and interpretable CNN–LSTM model for arrhythmia classification. The 11-layer architecture combines CNNs for spatial feature extraction with LSTMs for temporal pattern recognition and achieves an average accuracy of 98.24%. Its compact design makes it well-suited for deployment on portable cardiac monitoring devices. Overall, while CNN–LSTM hybrid models have demonstrated improvements in capturing both spatial and temporal ECG features, their classification accuracy—especially in multi-label settings—remains relatively limited and requires further enhancement for broader clinical adoption.

In the field of CNN–Transformer integration, several studies have achieved notable results. Naidji et al. [35] proposed a hybrid deep learning model that combines a Vision Transformer (ViT) with a Convolutional Neural Network (CNN) for detecting COVID-19 from ECG images. The model utilizes CNN layers to extract local features and ViT components to capture global dependencies. It achieves a high average accuracy of 98.78%, outperforming several state-of-the-art methods in ECG image classification. However, the study is limited by the relatively small dataset size and a potential risk of overfitting, indicating the need for further validation on larger and more diverse datasets. Khalid et al. [36] introduced another hybrid model that integrates CNNs with ViT for 12-lead ECG image classification. Their approach achieved an accuracy of 98.5% on a private dataset, demonstrating strong classification performance. Zhang et al. [37] presented a dual-branch hybrid architecture that combines CNN and Transformer modules for enhanced 12-lead ECG classification. One branch extracts local features using multi-scale CNNs, while the other captures global dependencies via a Transformer equipped with a token selection mechanism, which reduces redundancy and improves efficiency. The model was evaluated on the PTB-XL dataset, achieving an average AUC of 87.23 for the FORM task and 96.44 on the CPSC dataset. Although these CNN–Transformer hybrid models demonstrate improved classification accuracy over traditional approaches, their increased architectural complexity poses challenges for real-time or resource-constrained deployments, such as in wearable or IoT-based ECG monitoring systems.

3. Methodology

In this section, we introduce our method for multi-label ECG classification. We describe the insightful architecture of the proposed Inception Time Mamba Network, primarily constructed using the ITM Block. Implementation details of the two sub-modules, the Inception Block and Mamba Block, are also provided. Additionally, we discuss the application of the Distributed-Balanced Loss in our model.

3.1. Overview Framework

The illustration of our approach is in Fig. 1. The input to the model is a 12-lead ECG signal, represented as $12 \times L$, where L denotes the length of the signal. The first layer is a Convolutional Layer (Conv1D) followed by Batch Normalization (BN), which transforms the input to $C \times L$ where C is the projection dimension. This output is then passed through stages, with each stage consisting of ITM blocks (ITMBlock), which is detailed in the next section. The MaxPooling layer is placed at the end of each stage to reduce length. The first stage contains two ITM blocks that maintain the dimensions at $C \times L$, while the subsequent MaxPooling layer reduces the length to $C \times \frac{L}{2}$. This

Table 1

The configuration of the proposed structure. The terms K, S and Pool_K, Pool_S respectively represent the kernel and stride values of Convolution and MaxPooling layers.

Layer	Details	Feature Shape
Input	-	12x1000
Conv1D	K=1, S=1	64x1000
ITMBlock 1_1	K=1, S=1, Pool_K=3, Pool_S=1	64x1000
ITMBlock 1_2	K=1, S=1, Pool_K=3, Pool_S=1	64x1000
MaxPool	Pool_K=2, Pool_S=2	64x500
ITMBlock 2_1	K=1, S=1, Pool_K=3, Pool_S=1	64x500
ITMBlock 2_2	K=1, S=1, Pool_K=3, Pool_S=1	64x500
MaxPool	Pool_K=2, Pool_S=2	64x250
ITMBlock 3_1	K=1, S=1, Pool_K=3, Pool_S=1	128x250
GAP	-	128

process is repeated in the second stage, with two more ITM blocks and another MaxPooling layer, further reducing the dimensions to $C \times \frac{L}{4}$. In the final stage, the MaxPooling layer is not used, instead, only one ITM block is used and expands the channel of the output to $2C \times \frac{L}{4}$. After all stages, we use a Global Average Pooling (GAP) layer to aggregate and compress the features to $2C \times \frac{L}{4}$. The resulting feature map is then passed through a Linear layer, producing the final output, which is evaluated using Distribution-Balanced Loss [42]. The output is matched against the label to perform the desired classification. The configuration of the proposed model is detailed in Table 1.

3.2. Proposed ITM Block

The ITM Block, the central component of our architecture, is depicted in the middle of Fig. 2. This block begins with a Convolution layer, followed by a BatchNormalization layer, with the convolution kernel set to 1 for channel projection purposes. The output from the Convolution layer feeds into two distinct pathways: the Inception pathway and the Mamba pathway. The Inception pathway includes an Inception Time block, which is designed to retain multi-scale feature extraction capabilities. Meanwhile, the Mamba pathway features a Mamba block, a robust state-space model architecture that excels in sequence modeling. The settings of both the Inception Time Block and Mamba Block are finely adjusted to maintain a consistent feature shape across the pathways. Each pathway utilizes a ReLU activation function. The outputs of these two pathways are then merged through element-wise addition, aggregating the features. This dual-pathway approach enables the ITM Block to leverage the complementary strengths of both the Inception and Mamba strategies, extracting both local and global signal features to yield more robust and discriminative feature representations. Subsequent subsections will delve deeper into the functionalities of Inception Time and Mamba blocks.

3.2.1. Inception Time Block

This efficient Inception Time block, positioned on the left side of Figure 2, is inspired by the InceptionV4 [38] and modified that showcase a parallelized architecture aimed at capturing multi-scale features by employing multiple convolution operations with varying kernel sizes. Consider $\mathbf{x} \in \mathbf{R}^{C \times L}$ as the input of the Inception Time Block where C is the channel and L is the length of features. The input $\mathbf{x}$ is passed through two branches. The first branch includes a MaxPooling layer followed by a Convolution layer with a kernel size of 1. Unlike the MaxPooling layer in the general architecture of the ITMM that uses kernel = 2 and stride = 2 to reduce the length, the MaxPooling class in the Inception Time block uses kernel = 3 and stride = 1 and does not change the length of the input, as shown in Table 1. The output $\mathbf{x}_{pool}$ of the MaxPooling layer is then moved to the Convolution layer, which projects the input channel to a one-fourth of the original size and obtains the $\mathbf{x}_1$. The second branch starts with a bottleneck Convolution layer, which has a kernel size of 1 and takes the role of reducing the dimension size to one-fourth of the original value. The output $\mathbf{x}_{bot}$ of this Convolution is passed through three parallelized Convolution layers with kernel size values of 9, 19, and 39, obtaining outputs $\mathbf{x}_2$, $\mathbf{x}_3$ and $\mathbf{x}_4$ respectively. The outputs of two branches, or in detail words, of four Convolution layers are then concatenated to obtain the final feature. With $\mathbf{x} \in \mathbf{R}^{C \times L}$ is the input and $d = \frac{C}{4}$, we obtain the final output $\mathbf{y} \in \mathbf{R}^{C \times L}$ of Inception Time Block by following steps:

$$\mathbf{x}_{pool} = \text{MaxPool}(\mathbf{x}) \in \mathbf{R}^{C \times L} \quad (1)$$

$$\mathbf{x}_{bot} = \text{Conv1D}_{bot}(\mathbf{x}) \in \mathbf{R}^{d \times L} \quad (2)$$

$$\mathbf{x}_1 = \text{Conv1D}_{k=1}(\mathbf{x}_{pool}) \in \mathbf{R}^{d \times L} \quad (3)$$

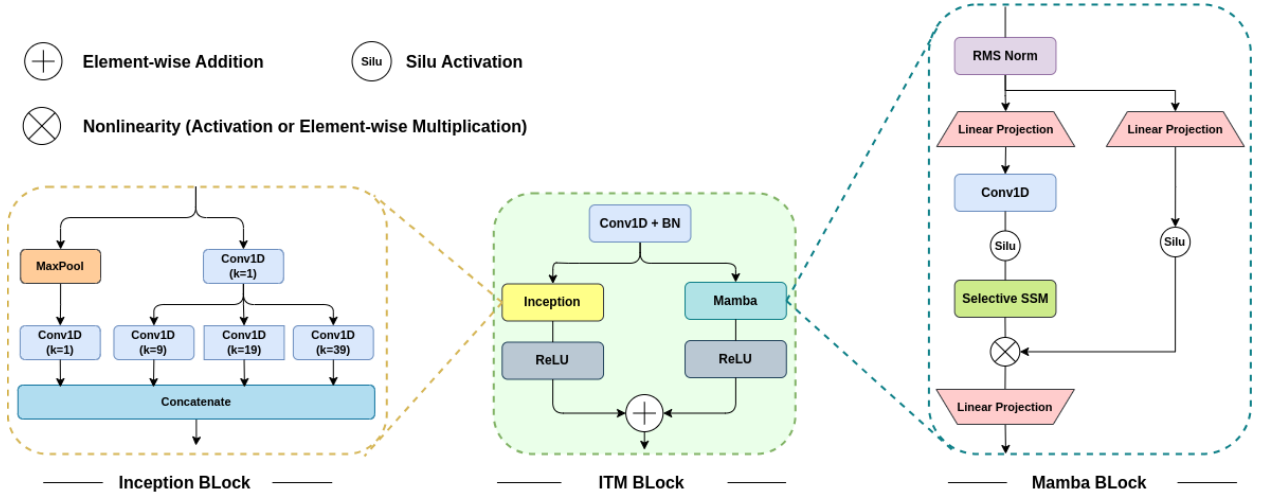


Figure 2: Illustration of our proposed ITM Block architecture (middle), highlighting its two sub-modules: the Inception Block (left) and the Mamba Block (right). The left branch focuses on extracting local ECG features, such as the positions of the P, T, and U waves, along with PR, QT, and ST intervals. The right branch captures global ECG characteristics, including QRS complex morphology, RR intervals, and wave amplitudes.

$$\mathbf{x}_2 = \text{Conv1D}_{k=9}(\mathbf{x}_{bot}) \in \mathbf{R}^{d \times L} \quad (4)$$

$$\mathbf{x}_3 = \text{Conv1D}_{k=19}(\mathbf{x}_{bot}) \in \mathbf{R}^{d \times L} \quad (5)$$

$$\mathbf{x}_4 = \text{Conv1D}_{k=39}(\mathbf{x}_{bot}) \in \mathbf{R}^{d \times L} \quad (6)$$

$$\mathbf{y} = \text{Concatenate}([\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4]) \in \mathbf{R}^{C \times L} \quad (7)$$

This approach enables the model to effectively capture local features at multiple scales, enriching the extracted representations. By combining outputs from convolutions with varying receptive fields, the Inception Time block reduces the risk of losing critical information that could occur with the use of a single convolutional operation.

3.2.2. Mamba Block

Before presenting Mamba Block, we first make a brief of Structure State Space Models (SSMs or S4) [43]. SSMs have emerged as a powerful model class for sequence modeling and time series data. SSMs are traditionally used in control theory to model a dynamic system via state variables and equations. Mathematically, at time t , SSMs map an input sequence $x(t) \in \mathbf{R}$ to a latent state representation $h(t) \in \mathbf{R}^N$ and derive a predicted output sequence $y(t) \in \mathbf{R}$ using first-order differential equations as:

$$h(t)' = \mathbf{A}h(t) + \mathbf{B}x(t), \quad (8)$$

$$y(t) = \mathbf{C}h(t) \quad (9)$$

where $\mathbf{A} \in \mathbf{R}^{N \times N}$, $\mathbf{B} \in \mathbf{R}^{N \times 1}$ and $\mathbf{C} \in \mathbf{R}^{1 \times N}$ are learnable matrices. The above equations stand for continuous SSMs. In practice, the SSMs can be discretized by a step size Δ , which can be seen as the resolution of the continuous input $x(t)$. The discretized SSMs are represented as:

$$h_t = \bar{\mathbf{A}}_{t-1} + \bar{\mathbf{B}}x_t, \quad (10)$$

$$y_t = \mathbf{C}h_t \quad (11)$$

where $\bar{\mathbf{A}} = \exp(\Delta\mathbf{A})$ and $\bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I})\Delta\mathbf{B}$.

Returning to the Mamba (or S6) model [39], it is currently the most advanced variant of state space models (SSMs). Mamba has demonstrated exceptional capabilities, quickly establishing itself as the state of the art (SOTA) in many sequence modeling challenges, even outperforming or competing with advanced Transformer variants [44].

Algorithm 1 The process of Mamba Block

Input: $\mathbf{x} \in \mathbf{R}^{B \times L \times D}$ **Output:** $\mathbf{y} \in \mathbf{R}^{B \times L \times D}$

```
1:  $\mathbf{x} = \text{RMSNorm}(\mathbf{x}) \in \mathbf{R}^{B \times L \times D}$ 
2: // Primary Pathway
3:  $\mathbf{x} = \text{Linear}(\mathbf{x}) \in \mathbf{R}^{B \times L \times ED}$ 
4:  $\mathbf{x}' = \text{SiLU}(\text{Conv1D}(\mathbf{x})) \in \mathbf{R}^{B \times L \times ED}$ 
5:  $\mathbf{A} = \text{Parameter}(N, N)$ 
6:  $\mathbf{B} = \text{Linear}(\mathbf{x}') \in \mathbf{R}^{B \times L \times N}$ 
7:  $\mathbf{C} = \text{Linear}(\mathbf{x}') \in \mathbf{R}^{B \times L \times N}$ 
8:  $\Delta = \text{Softplus}(\text{Parameter} + \text{Broadcast})(\text{Linear}(\mathbf{x}'))$ 
9:  $\bar{\mathbf{A}}, \bar{\mathbf{B}} = \text{discretize}(\Delta, \mathbf{A}, \mathbf{B})$ 
10:  $\mathbf{x}' = \text{SelectiveSSM}(\Delta, \bar{\mathbf{A}}, \bar{\mathbf{B}})(\mathbf{x}')$ 
11: // Second Pathway
12:  $\mathbf{z} = \text{SiLU}(\text{Linear}(\mathbf{x})) \in \mathbf{R}^{B \times L \times ED}$ 
13: // Merge and return output
14:  $\mathbf{y} = \mathbf{x}' * \mathbf{z} \in \mathbf{R}^{B \times L \times ED}$ 
15:  $\mathbf{y} = \text{Linear}(\mathbf{y}) \in \mathbf{R}^{B \times L \times D}$ 
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Unlike the Transformer's attention mechanism, which has quadratic computational complexity, Mamba achieves efficient computation with linear complexity. Traditional SSM models are often constrained by context-based reasoning challenges due to their linear time-invariant (LTI) property, meaning the model parameters do not change in response to the input sequence. Mamba addresses this limitation with its core feature, Selective SSM, which introduces a selection mechanism that adapts the parameters of the SSM based on the input, taking into account the sequence length and batch size. Specifically, given the complete input $\mathbf{x} \in \mathbf{R}^{B \times L \times D}$ where B is the batch size, L is the length (or number of variates) and D is the hidden dimension, the parameters $\mathbf{B}$, $\mathbf{C}$, Δ are dependent on the input $\mathbf{x}$ as:

$$\mathbf{B}, \mathbf{C}, \Delta = \text{Linear}(\mathbf{x}) \quad (12)$$

where $\mathbf{B} \in \mathbf{R}^{B \times L \times N}$, $\mathbf{C} \in \mathbf{R}^{B \times L \times N}$ and $\Delta \in \mathbf{R}^{B \times L \times D}$.

The Selective SSM can be efficiently computed by hardware-aware techniques implemented in [39]. In this study, we want to take advantage of Mamba, such as global receptive fields and dynamic weighting, to capture global features. The insight architecture of a vanilla Mamba Block is shown in the right part of Fig. 2. The Mamba Block starts with an RMS normalization layer, followed by two parallel pathways. Both pathways have a Linear Projection layer to expand the hidden dimension D to $E \cdot D$ where E is the expanding ratio. In the primary pathway, the output of the Linear Projection layer is sequentially passed through a Convolution layer, a SiLU activation function [45], and a Selective SSM module. In the second pathway, the output of the Linear Projection layer is simply passed through a SiLU activation function. Then the outputs of the two pathways are now merged by the element-wise multiplication. Another Linear Projection layer is used to reduce the hidden dimension and obtain the final output. The Transpose operation (not shown for simplicity), is applied on the input and output of the Mamba block to ensure the feature shapes are suitable for computation with other modules. The process of Mamba Block is detailed in Algorithm 1 to clarify the flow of Mamba as well as the calculation steps in equations (8) to (12).

3.3. Propose to use Distribution-Balanced Loss

One of the major problems affecting ECG classification results is the data imbalance. In practice, the amount of data belonging to some classes (e.g., healthy, normal heart rate) is much more significant than other classes (ill heart rate), leading to long-tailed datasets. Physically collecting data for minor classes requires additional resources. Using a generative model is one of the most effective approaches. Several generator models [46] demonstrate the ability to generate realistic ECG signals. However, this method requires additional costs for generation and verification, especially when the number of classes to be classified increases. We optimally approach this problem, using the data itself instead of an external resource. More specifically, we use the Distribution-Balanced (DB) Loss function [42] to train the classification model. In the scenario of multi-label classification, Binary Cross Entropy (BCE) loss is

the standard choice. Nonetheless, the plain BCE is proven to be vulnerable to label imbalance [47]. Meanwhile, the Distribution-Balanced Loss is a loss function proposed to address multi-label recognition problems that exhibit long-tailed class distributions. The efficiency of DB Loss is proven via remarkable results on many long-tailed datasets in different domains. Given the dataset $\mathcal{D} = \{(\mathbf{x}^1, \mathbf{y}^1), \dots, (\mathbf{x}^N, \mathbf{y}^N)\}$ where N is the number of training samples. Each instance is a sample-label pair where label $\mathbf{y}^k = [y_1^k, y_2^k, \dots, y_C^k]$ (C is the number of classes). The DB Loss is constructed by two mechanisms: Re-balanced Weighting and Negative-Tolerant Regularization (NTR).

3.3.1. Re-balanced Weighting

This mechanism takes the role of re-balancing the weights by adjusting the sampling probabilities to account for the imbalance introduced by label co-occurrence. The re-balanced weight r_i^k for each instance $\mathbf{x}^k$ and class i is computed as $r_i^k = \frac{P_C^i(\mathbf{x}^k)}{P_I(\mathbf{x}^k)}$, where $P_C^i(\mathbf{x}^k)$ is the class-level sampling frequency and $P_I(\mathbf{x}^k)$ is the instance-level sampling frequency. This re-balanced weight is then incorporated with a smoothing function to obtain the smoothed re-balanced weight, formulated as $\hat{r}_i^k = \alpha + \frac{1}{1 + \exp(-\beta \cdot (r_i^k - \mu))}$ where α , β and μ are hyperparameters. The re-balanced binary cross-entropy (R-BCE) loss is defined as:

$$\mathcal{L}_{R-BCE}(x^k, y^k) = \frac{1}{C} \sum_{i=1}^C [y_i^k \log(1 + e^{-z_i^k}) + (1 - y_i^k) \log(1 + e^{z_i^k})] \times \hat{r}_i^k \quad (13)$$

where z^k denotes the output of the classifier.

3.3.2. Negative-Tolerant Regularization (NTR)

This mechanism helps to treat the positive and negative instances of the same label differently. More specifically, the NTR addresses the over-suppression of negative labels by introducing a margin v_i and a scaling factor λ to the negative logits, resulting in the Negative-Tolerant-BCE (NT-BCE) loss as:

$$\mathcal{L}_{NT-BCE}(x^k, y^k) = \frac{1}{C} \sum_{i=1}^C [y_i^k \log(1 + e^{-(z_i^k - v_i)}) + \frac{1}{\lambda} (1 - y_i^k) \log(1 + e^{\lambda(z_i^k - v_i)})] \quad (14)$$

Finally, by combining the R-BCE loss and NT-BCE loss, the Distribution-Balanced Loss is defined as:

$$L_{DB}(x^k, y^k) = \frac{1}{C} \sum_{i=1}^C \hat{r}_i^k [y_i^k \log(1 + e^{-(z_i^k - v_i)}) + \frac{1}{\lambda} (1 - y_i^k) \log(1 + e^{\lambda(z_i^k - v_i)})] \quad (15)$$

By leveraging the benefits of the above mechanisms, the Distribution-Balanced Loss helps to smooth the distribution of the classifier outputs, especially for those tail classes. This loss can be integrated for end-to-end training as other standard loss functions such as Binary Cross Entropy loss.

4. Experiments

In this section, we present two datasets employed in the experiments, outline the evaluation metrics for multi-label classification, and describe the experimental implementations.

4.1. Datasets

We evaluate our approach on two standard public ECG detection datasets, the Physikalisch Technische Bundesanstalt Extra Large (PTB-XL) dataset [48] and the China Physiological Signal Challenge 2018 (CPSC) dataset [49], respectively.

4.1.1. PTB-XL Dataset

The PTB-XL dataset is a large 12-lead ECG dataset with 21,837 clinical ECG recordings from 18,885 patients. We have removed samples that missing labels, leaving 21,799 samples. Each ECG record contains an ECG signal of 10 seconds, along with multiple ECG statements (i.e., multiple labels) annotated by up to two ECG experts. The ECG signals are sampled at 100 Hz and 500 Hz, and we use the 100 Hz data in this work. The PTB-XL dataset

Table 2

Number of ECG samples for training, validation, and testing sets in different tasks of PTB-XL dataset and CPSC dataset.

Task	# Classes	# Train	# Validation	# Test	# Total
all	71	17418	2183	2198	21799
diag	44	17084	2146	2158	21388
sub-diag	23	17084	2147	2159	21390
super-diag	5	17084	2148	2160	21392
form	19	7197	901	880	8978
rhythm	12	16831	2100	2098	21029
CPSC	9	5501	687	689	6877

has a total of 71 different ECG statements. These statements can be grouped into 44 diagnostic statements (diag.), 19 formal statements (form), and 12 rhythmic statements (rhythm). The diagnostic statements are further divided into two subgroups: 24 subclasses (sub-diag.) and 5 coarse superclasses (super-diag.) according to their hierarchical structure. We do experiments on the six types of statements mentioned above. We follow the dataset splitting procedure recommended in the original PTB-XL paper, dividing the data into ten balanced folds based on the class distribution. Specifically, folds 1–8 constitute the training set, fold 9 serves as the validation set, and fold 10 is designated as the test set. This approach ensures that ECG records from each patient remain within a single fold, effectively preventing data leakage caused by patient data appearing in multiple folds. Before training and testing, ECG signals are preprocessed using a 4th-order Butterworth bandpass filter with cutoff frequencies of 0.4 Hz (low cutoff) and 45 Hz (high cutoff) to remove baseline wander and powerline interference. Table 2 summarizes the sample counts for the training, validation, and test sets corresponding to each task.

4.1.2. CPSC 2018 Dataset

The CPSC 2018 dataset, also known as the China Physiological Signal Challenge 2018 dataset, is a comprehensive collection of electrocardiogram (ECG) recordings designed for the detection and classification of arrhythmias. This dataset contains 6877 12-lead ECG recordings ranging from 6 to 60 seconds. These recordings were obtained from patients with various cardiac conditions and included annotations for 9 different types of arrhythmias. To standardize the data, we first reduce the sampling frequency from the original 500Hz to 100Hz. Then recordings longer than 10 seconds were trimmed, whereas those shorter were extended to reach a uniform duration of 10 seconds. Finally, the dataset was partitioned into ten groups. Specifically, group 9 was designated as the validation set, group 10 as the test set, and the remaining groups (1–8) were used for training. Prior to model training and evaluation, ECG signals were preprocessed to eliminate baseline wander and powerline interference using a 4th-order Butterworth bandpass filter with cutoff frequencies of 0.4 Hz and 45 Hz. The distribution of samples across the training, validation, and test sets for each specific task is detailed in Table 2.

4.2. Evaluation Metrics

To address the issue of category imbalance in our datasets, we employ the macro-AUC score as our primary evaluation metric, due to its robustness against class imbalance—unlike the traditional AUC, which can be overly influenced by majority classes. The AUC, which measures the Area Under the Receiver Operating Characteristic (ROC) Curve, plots the False Positive Rate (FPR) on the x-axis and the True Positive Rate (TPR)—also known as Sensitivity (SEN)—on the y-axis. While the AUC provides an overall view of the classifier’s performance, it can sometimes mask deficiencies in minority class detection, particularly in imbalanced settings. To better reflect performance on minority classes, we also report the F1-score, defined respectively as:

$$\text{FPR} = \frac{FP}{FP + TN}, \quad \text{TPR} = \frac{TP}{TP + FN}, \quad \text{Precision} = \frac{TP}{TP + FP}, \quad F_1 = 2 \cdot \frac{\text{Precision} \times \text{TPR}}{\text{Precision} + \text{TPR}}$$

In this context, FP, TN, TP, and FN represent the counts of false positives, true negatives, true positives, and false negatives, respectively.

Furthermore, to evaluate performance across multiple classes or detection scenarios, we calculate the mean Average Precision (mAP) as:

$$\text{mAP} = \frac{1}{N} \sum_{i=1}^N \text{AP}_i$$

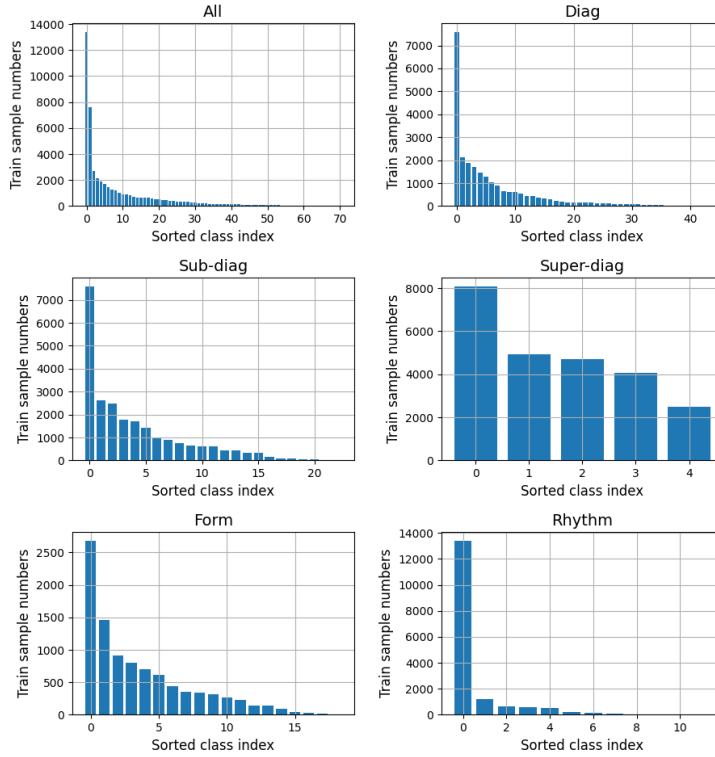


Figure 3: The distribution of classes in the PTB-XL dataset —categorized into All, Diag, Sub-diag, Super-diag, Form, and Rhythm—corresponds to different classification tasks. In the 'All' category, the difference in sample numbers between the most and least represented classes is approximately 280 times. For the 'Diag' and 'Super-diag' categories, this difference is approximately 76 times, whereas for the 'Sub-diag' category, it is about 3.8 times. The 'Form' category exhibits a class imbalance of roughly 22 times. Notably, the 'Rhythm' category shows the greatest imbalance, nearly 280 times, similar to the 'All' category; however, in the 'Rhythm' category, most samples are concentrated within a few dominant classes.

where N is the total number of classes. AP_i is the average precision for class i .

4.3. Implementation Details

Training is conducted in mini-batches of 64 for a total of 50 epochs, using the AdamW optimizer with a weight decay of 0.0001. The initial learning rate is set at 0.001 and is reduced by a factor of 10 after the 15th epoch. For loss computation, we employ the standard binary-cross entropy loss to facilitate multi-label classification. In experiments utilizing the Distribution-Balanced (DB) loss, we calculate the sampling frequency based on the number of samples per class in the training dataset. Figure 3 shows the class distribution in the training set for various classification tasks of the PTB-XL dataset, emphasizing the data imbalance. The hyperparameters for DB loss, designated as α , β , and μ , are set to values of 0.1, 10, and 0.05, respectively. All hyper parameters and selected values in the method are tuned and tested many times to give the best overall performance. All experiments are conducted using the PyTorch framework on a single RTX 3090 GPU. The code for our study is available at <https://github.com/tnquoc/ITMN>.

5. Results and Discussion

This section presents the results of our experiments. Additionally, we conduct ablation studies to discuss and analyze the impact of each component of our proposal.

5.1. Comparison with Current Cutting-Edge Research

Table 3 shows the performance comparison between our proposal and various models on the PTB-XL dataset, where the best and second best results for each task are bolded and underlined respectively. The results indicate that our full setting model, ITM + DB Loss exhibits superior performance by having the highest AUC and TPR values on

Table 3

Performance comparison of various models on PTB-XL dataset.

Model	all		diag		sub-diag		super-diag		form		rhythm	
	AUC	TPR	AUC	TPR	AUC	TPR	AUC	TPR	AUC	TPR	AUC	TPR
ACNet [31]	89.54	67.81	89.51	66.27	89.69	67.45	92.33	78.14	83.12	47.65	95.9	77.82
ATI-CNN [9]	89.51	70.23	90.83	67.17	90.8	70.34	91.81	77.77	83.54	55.58	96.69	91.17
LSTM [25]	92.74	67.97	91.72	64.28	91.66	67.36	92.44	76.21	85.14	48.82	91.27	87.19
BiLSTM [26]	91.35	69.53	91.37	64.13	91.98	69.13	92.19	78.61	82.14	49.24	94.93	88.42
FCN_Wang [40]	91.24	68.88	91.97	65.63	90.15	69.9	92.02	76.42	85.43	49.42	92.61	88.37
Resnet1D_Wang [40]	91.13	69.32	92.12	66.44	90.91	68.23	92.17	75.43	86.04	52.97	95.06	89.71
XResnet1D-101 [41]	90.61	68.89	91.42	64.69	89.98	67.95	91.95	75.27	82.02	51.35	94.9	91.15
MiniRocket [50]	60.38	67.21	57.24	67.41	62.06	69.31	73	69.45	54.05	60.72	51.71	62.96
MobilenetV3 [51]	90.12	68.81	89.53	61.34	89.25	64.06	91.29	75.4	81.8	48.25	96.22	91.39
InceptionTime [38]	92.11	71.93	92.82	66.24	92.39	70.95	92.61	78.17	86.66	59.05	95.03	90.11
ViT [52]	78.23	57.69	81.73	48.63	83.04	51.83	81.64	61.49	71.61	21.43	77.37	81.26
MVMS [17]	92.89	<u>73.66</u>	94.35	<u>68.97</u>	93.7	<u>74.21</u>	<u>93.25</u>	<u>81.29</u>	89.1	56.16	95.65	<u>91.83</u>
ITM	<u>96.22</u>	71.49	92.75	67.39	<u>95.68</u>	69.17	92.78	79.09	<u>91.28</u>	53.43	97.35	90.06
ITM + DB Loss	97.63	75.22	95.64	72.11	96.65	76.63	93.64	81.44	91.75	<u>58.74</u>	<u>97.08</u>	92.37

Table 4

Performance comparison of different models on CPSC and PTB-XL (task Form) datasets, using all evaluation metrics.

Method	CPSC				PTB_XL (form)			
	AUC	TPR	F1	mAP	AUC	TPR	F1	mAP
ACNet [31]	94.66	80.94	58.15	83.62	83.12	47.65	15.46	64.96
ATI-CNN [9]	94.73	83.26	76.63	87.04	83.54	55.58	23.22	68.43
LSTM [25]	94.81	74.93	72.41	87.56	85.14	48.82	22.13	70.84
BiLSTM [26]	95.06	77.75	76.65	89.19	82.14	49.24	23.5	69.42
FCN_Wang [40]	91.85	70.59	69.8	84.36	85.43	49.42	23.46	69.56
Resnet1D_Wang [40]	94.72	76.74	75.52	87.28	86.04	52.97	29.75	71.01
XResnet1D-101 [41]	95.22	82.61	79.78	90.6	82.02	51.35	31.12	70.16
MiniRocket [50]	73.23	60.49	62.39	76.22	54.05	60.72	19.03	50.11
MobilenetV3 [51]	95.23	81.52	75.64	85.91	81.8	48.25	20.07	49.63
InceptionTime [38]	94.47	79.28	76.84	87.33	86.66	59.05	32.32	72.37
ViT [52]	82.58	47.1	64.89	78.46	71.61	21.43	18.98	61.2
ECGNet [27]	95.11	82.68	79.49	88.31	83.3	54.89	27.24	54.89
CNN_Transformer [37]	<u>96.44</u>	87.32	<u>84.62</u>	<u>91.03</u>	87.23	68.26	<u>33.45</u>	<u>74.59</u>
MVMS [17]	95.8	84.98	75	86.85	89.1	56.16	31.63	72.93
ITM	94.89	84.18	77.03	87.43	<u>91.28</u>	53.53	31.27	70.66
ITM + DB Loss	96.68	<u>85.34</u>	85.49	91.81	91.75	<u>58.74</u>	33.59	74.95

most tasks (except TPR in task "form" and AUC in task "rhythm"). In comparison to the current SOTA method, MVMS, our model demonstrates an average Area Under the Curve (AUC) that is 2.24% higher, with task-specific differences of 4.74%, 1.3%, 2.95%, 0.39%, 2.65%, and 1.43%, respectively. Additionally, our model exhibits an improved average TPR, with an overall increase of 1.74%, and task-specific variations of 1.56%, 3.24%, 2.42%, 0.15%, 2.58%, and 0.54%. Figure 4(a) and Figure 4(b) respectively visualize comparing the two metrics AUC and TPR between our proposals and the SOTA MVMS model on different classification tasks.

Table 4 presents a comparative evaluation of the proposed ITM + DB Loss model against previous state-of-the-art methods, namely CNN_Transformer[37] and MVMS[17], using standard evaluation metrics including AUC, TPR, F1-score, and mAP. The experiments were conducted on the CPSC and PTB-XL (form task) datasets. The results demonstrate that our proposed model consistently outperforms both baselines across all key metrics. On the CPSC dataset, ITM + DB Loss achieves an AUC of 96.68%, which is 0.24% higher than CNN_Transformer (96.44%), along with improvements of 0.97% in F1-score and 0.59% in mAP. Compared to MVMS, the gains are even more substantial: 0.88% higher in AUC, 0.36% in TPR, 10.49% in F1-score, and 4.96% in mAP. On the PTB-XL dataset, the performance margins become more significant. The proposed model surpasses CNN_Transformer [37] by 4.52% in AUC and shows notable gains in both F1-score and mAP. Compared to MVMS, ITM + DB Loss achieves improvements of 2.65% in

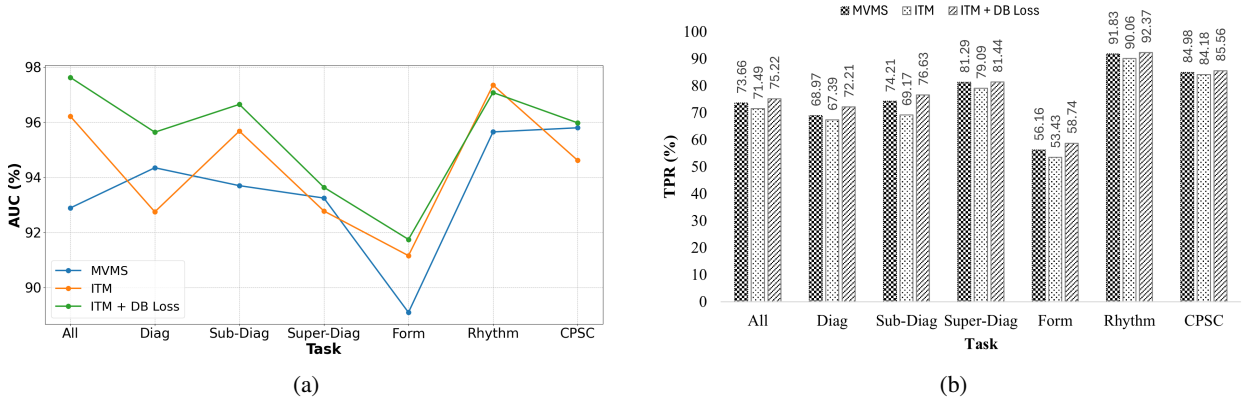


Figure 4: Comparison of AUC (a) and TPR (b) among seven groups of experiments on PTB-XL and CPSC datasets.

Table 5

The number of parameters (Params), FLOPs value, and inference time (Infer) for each model. We compute the inference time for 64 ECG samples on the PTB-XL dataset. The ms stands for milliseconds.

Model	Params (10^6)	FLOPs (10^6)	Infer (ms)
FCN_Wang [40]	0.28	137.91	4.37
Resnet1D_Wang [40]	0.29	16.64	1.7
Xresnet1d-101 [41]	1.53	69.73	19.08
InceptionTime [38]	0.47	237.58	15.13
LSTM [25]	0.81	404.10	35.18
BiLSTM [26]	2.14	2140.6	70.51
ViT [52]	9.34	95.54	3.21
MobileNetV3 [51]	1.48	10.16	9.04
ACNet [31]	0.03	2.66	3.55
ATI-CNN [9]	5.00	287.34	13.52
ECGNet [27]	0.34	167.9	25.24
CNN_Transformer [37]	26.5	1413.9	19.62
MVMS [17]	0.39	82.27	8.91
Ours	0.43	46.95	7.05

AUC, 2.44% in TPR, 1.96% in F1-score, and 2.02% in mAP. Although the TPR of ITM + DB Loss on PTB-XL (58.74%) is slightly lower than that of CNN_Transformer (68.26%), the overall results clearly indicate superior performance across the remaining critical metrics. These consistent improvements across all major evaluation metrics highlight the superior generalization capability and robustness of the proposed ITM + DB Loss model, particularly in addressing the challenges of complex, multi-label ECG classification tasks.

As shown in Table 5, our model achieves an excellent trade-off between efficiency and complexity, with only 0.43M parameters, 46.95M FLOPs, and an inference time of 7.05 ms for processing 64 ECG samples. Compared to transformer-based models such as ViT [52] and CNN_Transformer [37], which require significantly more parameters (9.34M and 26.5M, respectively) and FLOPs (95.54M and 1413.9M), our model is up to 2.8× faster while being substantially more lightweight. Similarly, MVMS [17] has a comparable parameter size (0.39M) but incurs 75% more FLOPs and 21% slower inference (8.91 ms vs. 7.05 ms), indicating better efficiency of our approach. While ultra-lightweight models like ResNet1d14 [40] and ACNet [31] achieve faster inference (1.7 ms and 3.55 ms, respectively), they sacrifice modeling capacity. For instance, ACNet operates with only 2.66M FLOPs and achieves around 80% sensitivity, whereas our model attains 85.34%, demonstrating superior clinical reliability. In addition, models with high complexity such as BiLSTM [26], LSTM [25], ECGNet [27], and Xresnet1d101 [41] suffer from long inference times (e.g., up to 70.51 ms for BiLSTM), making them impractical for real-time or embedded applications. In summary, our model balances performance and efficiency effectively, making it well-suited for real-time ECG classification in resource-constrained IoT environments.

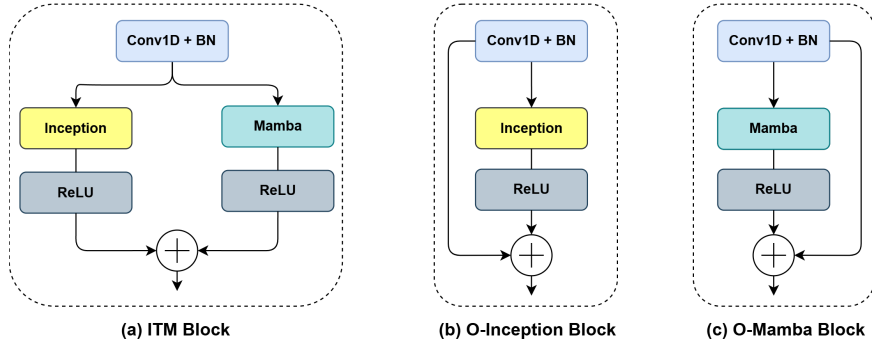


Figure 5: Illustration of (a) the ITM architecture, (b) the O-Inception topology, and (c) the O-Mamba block, respectively, utilized during training and validation to demonstrate the effectiveness of combining Inception and Mamba blocks for multi-label, multi-class ECG signal classification.

5.2. Impact of components in the proposed ITM block

We aim to explore the individual effects of the components within the ITM block, specifically the Inception block and the Mamba block. To accomplish this, we divide the ITM block into two separate sub-blocks, each containing either the Inception Block or the Mamba Block within a residual structure, respectively named O-Inception (Only Inception) and O-Mamba (Only Mamba). Figure 5 illustrates the architecture of the ITM block alongside the O-Inception and O-Mamba blocks. The objective of this experiment is to evaluate the influence of local and global features. It is important to note that this division effectively creates two new model architectures, each with unique structures that may yield different results. To ensure stability, the O-Inception block and O-Mamba block will replace the ITM block in the original model architecture as depicted in Figure 1. We will train these two architectures both with and without DB Loss to assess their performance.

Table 6 details the performance of the above architectures across various tasks, with and without Distribution-Balanced (DB) Loss, using AUC and TPR as metrics. Without DB Loss, O-Inception achieves a marginally higher overall AUC compared to O-Mamba, outperforming in 4 out of 6 tasks ('sub-diag', 'super-diag', 'form', 'rhythm') in the PTB-XL set. Regarding TPR, O-Inception consistently excels, showing higher TPR values in 4 out of 6 tasks, except the 'sub-diag' and 'super-diag' tasks. On the CPSC set, the two approaches are balanced: O-Inception has a lower AUC but a higher TPR. Overall, O-Inception demonstrates slightly better performance than O-Mamba. When using DB-Loss, all the metrics of O-Inception and O-Mamba increased significantly in all tasks. O-Inception catches up and surpasses O-Mamba in AUC in all tasks, particularly in tasks like 'all', 'diag', 'form', and 'rhythm' with a large margin. O-Mamba delivers better TPR results in 4 out of 6 tasks ('all', 'diag', 'sub-diag', and 'super-diag'), though the difference from O-Inception is not substantial. However, O-Mamba struggles significantly with the 'form' and 'rhythm' tasks, regardless of the presence of DB Loss. On the CPSC set, both models exhibit similar AUC values, with O-Inception achieving a slightly higher TPR. Both sub-modules benefit from DB Loss, but O-Inception experiences more pronounced improvements. With or without DB Loss, O-Inception is consistently comparable to or better than O-Mamba in several tasks. This is particularly clear when comparing O-Inception and O-Mamba to the ITM model. O-Inception's performance is closer to that of the ITM model than O-Mamba's. The findings suggest that the combination of Inception and Mamba is effective, as the ITM model surpasses both O-Inception and O-Mamba individually across most metrics in all settings. More specifically, ITM outperforms O-Inception with the average AUC and TPR differences of 3.38% and 1.27%, respectively, while with O-Mamba it is 4.62% and 2.93%. However, it appears that local features from O-Inception have a more significant impact on ECG classification than the global features from O-Mamba.

5.3. Impact of Distribution-Balanced Loss

The results from previous experiments demonstrate that Distribution-Balanced Loss (DBL) significantly enhances performance by addressing data imbalance effectively. In this section, we aim to further investigate the impact of different loss functions on handling data imbalance. Specifically, we will compare DB Loss with other popular loss functions tailored for this issue, including Weighted Binary Cross Entropy (WBCE), Focal Loss (FL) [53], and Class Balanced Focal Loss (CBFL) [54]. WBCE represents a traditional approach where standard Binary Cross Entropy

Table 6

Performance of each sub-module within ITM Block.

Methods	Metrics	Tasks						
		all	diag	sub	super	form	rhythm	CPSC
Without DB Loss								
O-Inception	AUC	85.29	89.71	95.64	92.78	75.09	86.86	93.01
	TPR	70.86	66.27	68.49	77.64	52.66	89.31	82.51
O-Mamba	AUC	86.89	91.61	95.39	92.59	62.68	64.55	95.59
	TPR	69.52	65.92	68.87	78.19	46.46	88.66	78.08
ITM	AUC	96.22	92.75	95.68	93.63	91.28	97.35	94.62
	TPR	71.49	67.39	69.17	79.09	53.43	90.06	84.18
With DB Loss								
O-Inception	AUC	97.22	95.13	95.07	92.87	80.09	89.81	93.89
	TPR	74.49	72.11	73.71	80.75	54.48	91.61	82.87
O-Mamba	AUC	95.07	92.58	94.78	92.25	70.92	71.23	93.57
	TPR	74.52	72.42	74.46	80.94	52.05	90.28	81.08
ITM	AUC	97.63	95.64	96.65	93.64	91.75	97.08	95.98
	TPR	75.22	73.13	76.63	81.44	58.74	92.37	85.34

Table 7

Performance of ITMN with different loss functions on classification tasks of PTB-XL dataset.

Task	Metric	Loss Function				
		BCE	WBCE	FL	CBFL	DBL
all	AUC	96.22	97.43	96.41	97.13	97.63
	TPR	71.49	71.95	74.45	74.79	75.22
diag	AUC	92.75	94.85	95.12	95.36	95.64
	TPR	67.39	67.72	71.88	72.04	72.11
sub-diag	AUC	95.68	96.22	95.38	96.63	96.65
	TPR	69.17	69.36	75.25	75.38	76.63
super-diag	AUC	92.78	93.11	93.05	93.35	93.64
	TPR	79.09	79.67	78.38	79.72	81.44
form	AUC	92.83	92.97	93.22	94.56	91.75
	TPR	53.43	53.97	55.93	56.26	58.74
rhythm	AUC	97.35	96.92	97.37	97.81	97.08
	TPR	90.06	90.18	90.91	91.12	92.37

Note: **BCE**: Binary Cross Entropy, **WBCE**: Weighted Binary Cross Entropy, **FL**: Focal Loss, **CBFL**: Class Balanced Focal Loss, **DBL**: Distribution Balanced Loss.

(BCE) is modified by applying weights to each class. We adopt the Class-Balanced weighting method from [54], which efficiently calculates weights based on the number of samples per class. Focal Loss (FL) [53], initially designed for object detection, focuses on enhancing the model's ability to learn from hard-to-classify examples, particularly useful in scenarios with significant class imbalance. Class Balanced Focal Loss (CBFL) [54] merges class-balanced weighting with Focal Loss. According to findings in [54], Class Balanced Focal Loss shows superior performance compared to Focal Loss alone, due to its dual focus on balancing class representation and prioritizing difficult examples. All loss functions are evaluated using their default hyperparameters to ensure a fair comparison and to reflect their typical implementation scenarios.

Table 7 provides a comprehensive comparison of ITMN's performance using various loss functions on classification tasks within the PTB-XL dataset, evaluated using AUC and TPR metrics. Overall, it can be observed that loss functions designed to handle data imbalance yield better results than the standard Binary Cross-Entropy (BCE) loss. Among them, the Distribution-Balanced Loss (DBL) consistently outperforms others, particularly excelling in the "all" task, achieving the highest AUC of 97.63% and TPR of 75.22%, as well as in the "diag" task with an AUC of 95.64% and TPR of 72.11%. In the "sub-diag" task, DBL achieves an AUC of 96.65% and a TPR of 76.63%, while in the "super-diag" task, it demonstrates superior performance with an AUC of 93.64% and a TPR of 81.44%. Although DBL

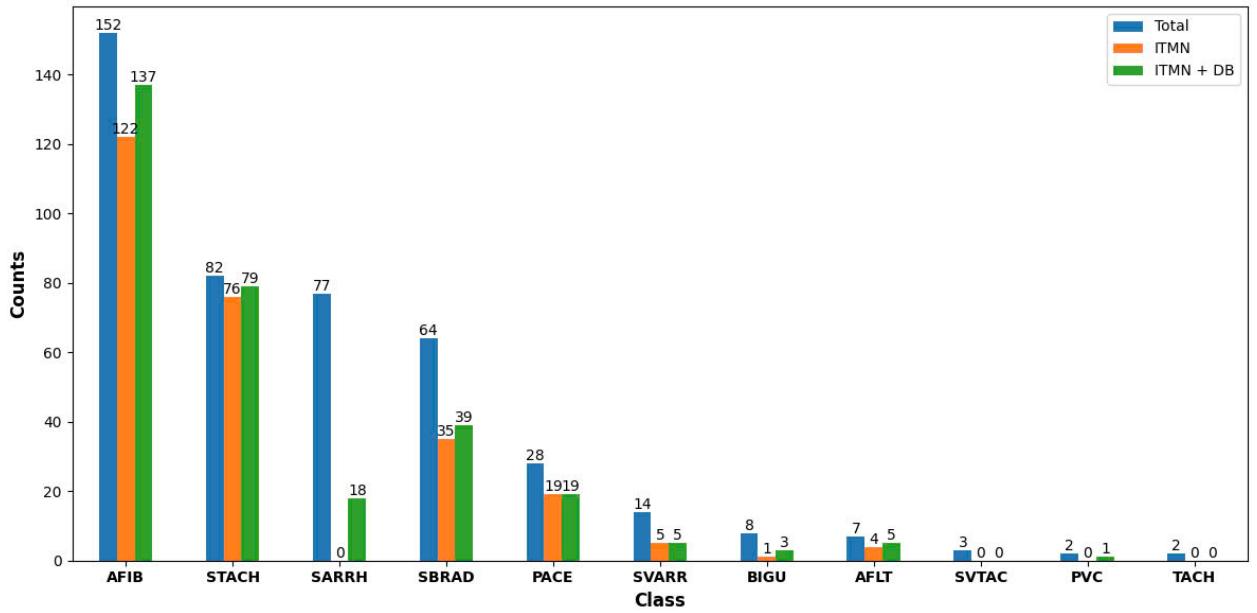


Figure 6: Comparison of True Positive (TP) counts for each class in the RHYTHM dataset using the ITMN model, with and without DB loss. Blue bars represent the total number of records per class, orange bars show TP values from the standard ITMN model, and green bars show TP values from ITMN with DB loss.

Table 8

The macro AUC scores of our method for each class on the SUPER task classification in the PTB-XL dataset.

Class	#Records	Description	AUC	
			ITMN	ITMN + DB Loss
NORM	9528	Normal ECG	94.46	94.83
CD	4907	Conduction Disturbance	92.07	92.31
MI	5486	Myocardial	89.34	90.03
HYP	2655	Hypertrophy	91.57	92.51
STTC	5250	ST/T Change	92.73	93.75

performs well in the “form” task with a TPR of 58.74%, its AUC of 91.75% is relatively lower, indicating potential for further improvement in this particular classification task. Notably, DBL achieves sub-optimal results in the “rhythm” task in terms of AUC (97.08%)—slightly behind CBFL—but still yields the highest TPR of 92.37%, underscoring its effectiveness in rhythm classification. Other loss functions, such as CBFL and FL, also demonstrate competitive performance. CBFL, in particular, may be considered a strong alternative due to its consistently high results across most tasks. However, DBL’s consistent top-tier performance across all task types highlights its strength in managing class imbalance and enhancing overall model accuracy. In summary, these results indicate that leveraging loss functions tailored for data imbalance is crucial for ECG multi-label classification, with the proposed Distribution-Balanced Loss emerging as the most stable and effective approach in this study.

5.4. Further Analysis of each class in datasets

We aim to further investigate the impact of incorporating DB loss, with a particular focus on how it affects performance across individual classes within various datasets. Given that some datasets contain a large number of classes (e.g., 71 in the “all” set and 41 in “diag”), we choose to analyze subsets with a more manageable number of classes—specifically, the SUPER dataset (5 classes), RHYTHM (12 classes), and CPSC 2018 (9 classes)—to save space and simplify analysis. Our evaluation compares the class-wise AUC macro scores of the ITMN model, both with and without the application of DB loss. Table 8 and Table 9 show the AUC macro scores for each class in the SUPER task of PTB-XL and the CPSC 2018 dataset, respectively. A consistent trend emerges: DB loss enhances performance

Table 9

The macro AUC scores of our method for each class on the CPSC 2018 dataset.

Class	#Records	Description	AUC	
			ITMN	ITMN + DB Loss
NORM	918	Normal ECG	97.22	97.74
AFIB	1098	Atrial Fibrillation	98.85	98.97
I-AVB	704	First-degree Atrioventricular Block	97.54	98.28
CLBBB	207	Left Bundle Branch Block	99.65	99.82
CRBBB	1695	Right Bundle Branch Block	98.78	98.89
PAC	574	Premature Atrial Contraction	89.72	90.21
VPC	653	Premature Ventricular Contraction	95.25	96.54
STD	826	ST-segment Depression	96.53	96.95
STE	202	ST-segment Elevated	93.24	93.78

across all classes. On average, it results in a 0.65% increase in the SUPER task and a 0.49% improvement in the CPSC 2018 task. These gains are reasonable, as the class imbalance in these datasets is moderate. Notably, even classes with the lowest sample ratios—such as MI in the SUPER task, or STE and CLBBB in the CPSC dataset—experience improvement with DB loss.

We further extend our analysis to the RHYTHM dataset, which exhibits the highest class imbalance (as illustrated in Figure 3). In this case, we calculate the number of True Positives (TP)—correctly predicted samples—for each class and compare them with the total number of test samples per class. Figure 6 illustrates that DB loss increases TP across all classes. A striking example is the SARRH class, where the model goes from 0 to 18 correct predictions when DB loss is applied. It also leads to improved TP counts in underrepresented classes such as BIGU, AFLT, and PVC. However, for extremely rare and complex classes like SVTACH and TACH, the model still fails to predict any samples correctly. Interestingly, DB loss also enhances performance in well-represented classes, such as AFIB (137/152) and STACH (79/82).

In summary, the results demonstrate that DB loss significantly enhances model performance on imbalanced datasets. It is particularly effective for handling rare ECG signals with very few available samples, making it valuable for real-world medical applications.

6. Conclusion

In this paper, we present the Inception Time Mamba Network (ITMN), a groundbreaking hybrid deep learning model poised to revolutionize multi-label classification of 12-lead ECG signals. ITMN's innovative fusion of Inception and Mamba blocks empowers it to expertly discern both subtle local patterns and overarching global trends, dramatically boosting noise robustness and feature richness. To overcome the persistent hurdle of class imbalance, we strategically employed a Distribution-Balanced loss function. Our rigorous experimental validation on two benchmark datasets unequivocally demonstrates ITMN's good performance over current state-of-the-art models. Crucially, this advanced capability comes packaged in a remarkably efficient architecture, boasting a mere 0.43 million parameters, making ITMN exceptionally well-suited for deployment in embedded IoT environments. While the promise of ITMN is clear, the critical next step involves validating its efficacy within the complexities of real-world clinical scenarios. Looking ahead, our research will explore integrating additional data modalities like ECG images and PPG signals, alongside applying transfer learning to pave the way for ubiquitous, real-time diagnostic support on wearable devices.

Acknowledgments

This research was supported by The VNUHCM-University of Information Technology's Scientific Research Support Fund. Thanks to the Computing Architecture Laboratory at the Nara Institute of Science and Technology in Japan for providing the facilities for this research.

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